adding the corresponding entries of both the matrices A and B.

Symbolically if $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{m\times n}$, then $A + B = [a_{ij} + b_{ij}]_{m\times n}$

Similarly $A - B = A + (-B) = [a_{ij}]_{m\times n} + [-b_{ij}]_{m\times n} = [a_{ij} - b_{ij}]_{m\times n}$

Multiplication of matrices

Multiplication of two matrices is possible only when the number of columns of the first matrix is equal to the number of rows of the second matrix. The product of matrices A and B is obtained by multiplying every row of matrix A with the corresponding elements of every column of matrix B element-wise and add the results.

Let $A = [a_{ij}]$ be an $m \times p$ matrix and $B = [b_{ij}]$ be a $p \times n$ matrix, then the product AB is a matrix $C = [c_{ij}]$ of order $m \times n$.

Transpose of a matrix

Let $A = [a_{ij}]$ be a matrix of order $m \times n$. The transpose of A, denoted by A^T of order $n \times m$ is obtained by interchanging either rows into columns or columns into rows of A.

For example,

if $A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 4 & 6 \end{bmatrix}$ then $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ 5 & 6 \end{bmatrix}$

NOTE

It is believed that the students might be familiar with the above concepts and our present syllabus continues from the following.

Definition 1.2

To every square matrix A of order n with entries as real or complex numbers, we can associate a number called ***determinant*** of matrix A and it is denoted by $|A|$ or det (A) or Δ.

Thus determinant can be formed by the elements of the matrix A.

If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then its

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

Example 1.1

Evaluate: $\begin{vmatrix} 2 & 4 \\ -1 & 4 \end{vmatrix}$

Solution

$$\begin{vmatrix} 2 & 4 \\ -1 & 4 \end{vmatrix} = (2)(4) - (-1)4$$

$$= 8+4 = 12$$

To evaluate the determinant of order 3 or more, we define minors and cofactors.

1.1.2 Minors:

Let $|A| = |[a_{ij}]|$ be a determinant of order n. The minor of an arbitrary element a_{ij} is the determinant obtained by deleting the ith row and j^{th} column in which the element a_{ij} stands. The minor of a_{ij} is denoted by M_{ij}.

1.1.3 Cofactors:

The cofactor is a signed minor. The cofactor of a_{ij} is denoted by A_{ij} and is defined as

$$A_{ij} = (-1)^{i+j} M_{ij}$$

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